Akashdeep Akashdeep

Experimental Physicist (PhD, Condensed Matter / Spintronics)

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LinkedIn | Google Scholar

Research profile snapshot

- **Research topics:** Altermagnetism in RuO_2; interfacial spin transport and magnetoresistance in oxide/ferromagnet heterostructures; spin dynamics in magnetic-insulator heterostructures (YIG/GdIG/Pt).
- **Core expertise:** Thin-film heterostructures; interface engineering; magnetotransport (low-T, high-B); advanced magnetic/structural characterization (sputter/PLD, XRD/XRR/AFM, SQUID/VSM, MOKE/FMR); large-scale methods (XMCD-PEEM, low-energy muon spectroscopy).
- **Publications & recognition:** First-author papers in Appl. Phys. Lett. (2026) and Phys. Rev. Applied (2025); publications in Science Advances, Phys. Rev. B, Advanced Materials, Applied Physics Letters; editorial selections in Phys. Rev. Applied and Appl. Phys. Lett.
- **Teaching & supervision:** TA for bachelor condensed matter physics; supervised bachelor project (RISE 2023–2024); hosted visiting PhD students and supported Master projects.

Research experience and projects

Ph.D. Researcher (Spintronics / Altermagnetism) — 2021 – Present

Johannes Gutenberg University Mainz, Mainz, Germany

- **Altermagnetism in RuO_2:** symmetry-breaking signatures in thin films/heterostructures; links between magnetic order and transport (Science Advances 2024; Phys. Rev. B 2026; arXiv submitted to *Nature Physics*)
- **Interfacial spin transport (RuO_2/Permalloy):** angle-/temperature-dependent magnetoresistance isolating interface-generated spin-current contributions (Phys. Rev. Applied 2025)
- **Depth-/surface-localized magnetism:** depth-resolved evidence for where magnetic order resides in RuO_2 thin films (Appl. Phys. Lett. 2026)
- **Beyond RuO_2:** imaging + transport support for altermagnetism in hematite and related systems (Advanced Materials 2025; arXiv submitted to *Nature Physics*)
- **Spin dynamics (YIG/GdIG/Pt):** contributions to temperature-dependent spin-dynamics studies in magnetic-insulator heterostructures (Phys. Rev. B 2025)

Junior Research Fellow (Thin Films / Magnetotransport) — 2020 – 2021

Indian Institute of Technology (IIT) Delhi, New Delhi, India

- Structure–property relationships in sputtered **Co_2MnAl** Heusler-alloy thin films (transport; magnetic anisotropy)

Education

Ph.D. in Physics (Experimental / Condensed Matter) — 2021 – expected Apr 2026

Johannes Gutenberg University Mainz (JGU), Mainz, Germany

Focus: spintronics, magnetoresistance devices, altermagnetism in RuO_2 and magnetic-insulator heterostructures
Supervisor: apl. Prof. Dr. Gerhard Jakob

M.Sc. in Physics — 2017 – 2019

Indian Institute of Technology (IIT) Delhi, New Delhi, India

Research project: *Sputtered Heusler-alloy Co_2MnAl thin films: transport and magnetic anisotropy*
Supervisor: Prof. Sujeet Chaudhary

B.Sc. in Physics — 2014 – 2017

University of Delhi, Ramjas College, Delhi, India

Experimental tools, methods, and technical skills

Thin-film synthesis & processing

- DC/RF magnetron sputtering
- PLD; thermal evaporation; spin coating
- Annealing; process optimization
- HV/UHV chamber integrations

Device fabrication (cleanroom)

- Optical & e-beam lithography
- Mask/layout design (KLayout)
- Lift-off; ion etching
- Wire-bonding

Characterization & measurement

- RHEED; XRD; XRR; AFM
- SQUID/VSM; MOKE; FMR
- Cryogenic magnetotransport (low-T, high-B)

Synchrotron & muon techniques

- XMCD-PEEM (BESSY II)
- Low-energy muon spectroscopy (SLS/PSI)

Software & data

- Python (analysis, plotting)
- LabVIEW (automation)
- OriginPro; LaTeX

Engineering / design tools

- SolidWorks (CAD)
- Inkscape (figures)

Conference contributions (talks and posters)

- MMM–Intermag, New Orleans, USA — **Talk** (Jan 2025)
- International Conference on Magnetic Films and Surfaces (ICMFS), Perugia, Italy — **Talk** (Jul 2024)
- DPG Spring Meeting (SKM), Regensburg, Germany — **Poster** (Mar 2022)
- International Conference on Advanced Materials, India — **Poster** (2019)

Teaching and supervision

- Teaching Assistant (Bachelor level): Condensed Matter Physics
- Supervision: Bachelor student project (RISE 2023–2024) — RT-FMR with modulating coil field
- Mentoring/support: hosted visiting PhD students (DGIST Korea; 2023, 3-month stays); supported Master projects (Permalloy thin-film characterization; rare-earth iron garnets YIG/TbIG)

Awards and major achievements

- Featured editorial selections in *Physical Review Applied* and *Applied Physics Letters*
- National competitive exams/rankings (India): AIR-35 (CSIR-NET 2019), AIR-107 (IIT-JAM 2017), AIR-125 (JEST 2017), AIR-706 (GATE 2019)
- Top 2% nationally in CBSE Higher Secondary examination
- National-level Gold medal (Netball)

Leadership and outreach

- Spin+X student supply-chain manager and speaker coordination
- Subgroup coordination/meetings (magnetic insulators/oxides/AFM)

References

Univ.-Prof. Dr. Mathias Kläui

JGU Mainz; Research Group Head
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apl. Prof. Dr. Gerhard Jakob

JGU Mainz; PhD Supervisor
jakob@uni-mainz.de

Publications

Google Scholar

1. **Akashdeep, A.** *et al.*
Surface-localized magnetic order in RuO_2 thin films revealed by low-energy muon probes.
Applied Physics Letters **128**, 022406 (2026).

2. **Akashdeep, A.** *et al.*
Angle-dependent magnetoresistance induced by interface-generated spin current in RuO_2/permalloy heterostructures.
Physical Review Applied **24**, 054018 (2025).

3. **Akashdeep, A.** *et al.*
Single spin Hall anomalous Hall and Hanle effect in YIG/Ta and YIG/Pt bilayers.
Manuscript in preparation (2026).

4. Y. Lytvynenko, **Akashdeep, A.** *et al.*
Magnetic circular dichroism in core-level x-ray photoelectron spectroscopy of altermagnetic RuO_2 films.
Physical Review B **113**, 014403 (2026).

5. M. Loyal, **Akashdeep, A.** *et al.*
Emergence of a spin Hall topological Hall effect in the non-collinear phase of the ferrimagnetic insulator terbium-iron garnet.
arXiv:2602.11721 (2026).

6. O. Fedchenko, J. Minár, **Akashdeep, A.** *et al.*
Observation of time-reversal symmetry breaking in the band structure of altermagnetic RuO_2.
Science Advances **10**, eadj4883 (2024).

7. F. Fuhrmann, S. Becker, **Akashdeep, A.** *et al.*
Temperature-dependent study of the spin dynamics of coupled $Y_3Fe_5O_{12}$/$Gd_3Fe_5O_{12}$/Pt trilayers.
Physical Review B **112**, 064432 (2025).

8. E. Galindez-Ruales, R. Gonzalez-Hernandez, C. Schmitt, S. Das, F. Fuhrmann, A. Ross, E. Golias, **Akashdeep, A.** *et al.*
Revealing the altermagnetism in hematite via XMCD imaging and anomalous Hall electrical transport.
Advanced Materials **37**(41), e05019 (2025).

9. E. Galindez-Ruales, W. Yang, T. Dannegger, M. Kundu, J. Köhler, C. Schmitt, F. Fuhrmann, **Akashdeep, A.** *et al.*
Altermagnetic magnon transport in the d-wave altermagnet $LuFeO_3$.
arXiv:2508.14569 (2025). Submitted to *Nature Physics*.

10. M. Weber, S. Wust, L. Haag, **Akashdeep, A.** *et al.*
All optical excitation of spin polarization in d-wave altermagnets.
arXiv:2408.05187 (2024). Submitted to *Nature Physics*.